Estimated metadata load from serving

	One cluster (keys/s)		Kimi fleet (keys/s)	
Key granularity	Lookups	Writes	Lookups	Writes
256-token block	160–220	90	4.5K	1.9–2.7K
64-token page per GPU	5.0–7.0K	2.9–3.0K	145K	59–87K
64-token page per GPU per layer	400–560K	230–240K	11.6M	4.7–6.9M
Capacity of one master	5.1M	175K	5.1M	175K

Inputs (Mooncake, FAST '25)

- One cluster: Kimi's 1-hour traces, 16 nodes $\times$ 8 GPUs: 40–56K prompt tokens/s, 40–59% cached.
- Fleet: over 100B tokens/day, about 1.16M tokens/s.

Model

- Lookups $=$ prompt tokens/s $\div$ tokens per key.
- Writes $=$ (1 $-$ cached fraction) $\times$ lookups.
- At most 5 calls per request: 17–33 calls/s per cluster.

Result. At block or page granularity, one cluster needs under 0.2% of the master's lookup capacity and under 2% of its write capacity. Only one key per layer exceeds the write capacity.

Hourly (cluster) and daily (fleet) averages; peaks are higher. All tokens counted as prompt tokens (97–98% are). Page per GPU: SGLang HiCache with 8 GPUs; 80 layers as in LLaMA3-70B. Capacity: E1 on 16 cores, no network; writes fall to 37.5K keys/s with 128 storage clients upstream and hold at 0.9M patched (E2). The fleet columns assume one master for all of Kimi.